MARS

Machine Assets Reporting Standard

ROI Methodology, Insurance Framework & Implementation Guide

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DOCUMENT IDENTITY	
Document title	MARS v0.9 — ROI Methodology, Insurance Framework & Implementation Guide
Technical purpose	Define a neutral, open, cross-manufacturer schema and calculation framework for autonomous machine performance, ROI, and insurance underwriting
Plain English purpose	Give every robot operator, insurer, and manufacturer the same measuring stick — so everyone agrees on what a robot is worth and how well it is working
Status	Public Draft v0.9 — actively seeking technical review from robotics engineers, actuaries, and fleet operators
Intended audience	Robot fleet operators, insurance underwriters, RaaS providers, OEM engineers, standards reviewers
What to trust fully	Formula structure, variable definitions, calculation logic, ISO standard references
What is estimated	Benchmark thresholds (TSR, MTBI ranges), MRS weighting factors, depreciation rates — labeled throughout
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1. What Is MARS?

1.1 Plain English

Today, every robot reports performance differently — or not at all. A logistics manager at a BMW plant cannot compare the economic performance of a Figure 02 against a Boston Dynamics Atlas using any neutral, shared standard. An insurer pricing coverage for a humanoid robot has no auditable data record to underwrite against. A CFO approving a 10-robot fleet purchase has no way to calculate genuine ROI across manufacturers.

MARS is the shared measuring stick that does not exist yet. It is an open data standard that any robot — regardless of manufacturer — can report against, producing records structured specifically for ROI calculation, insurance underwriting, and cross-fleet benchmarking.

Think of it as CARFAX for autonomous machines. Just as a used car's CARFAX report follows it from first sale through every owner, a robot's MARS record follows it from deployment through its entire operational life — substantiating its performance history, informing resale value, and enabling parametric insurance that triggers automatically on data.

1.2 Technical Definition

MARS is a JSON Schema (draft-07) telemetry standard defining 19 structured objects that capture the operational, AI performance, safety, energy, maintenance, and financial data required to compute the AM-ROI formula, the MARS Risk Score (MRS), and RaaS SLA compliance status for any autonomous machine deployment.

The standard is designed to be: integration-path agnostic (ROS2, vendor API, direct SDK, or manual entry); deployment-type normalised (benchmarks adjust by environment class); auditable (SHA-256 hash chain on every record); and extensible (additional modules planned for agricultural, healthcare, and construction deployments).

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2. Core Variables Reference

Every formula in this document uses the following variables. This table is the single source of truth.

Variable	Full Name	Definition
`P`	Purchase Price	Full acquisition cost of robot: hardware + initial software + installation. Currency as contracted. This is the capital base for all depreciation and payback calculations.
`L`	Expected Lifespan	Manufacturer or operator-estimated operational lifespan in years at rated performance. Determines annual capital cost: P / L.
`H_op`	Operating Hours	Total hours robot is operational in the measurement period (uptime). Includes autonomous and human-assisted operation.
`H_auto`	Autonomous Hours	Hours of operation completed without any human intervention in the period.

H_assist	Assisted Hours	Hours requiring human physical intervention. H_auto + H_assist = H_op.
TSR	Task Success Ratio	tasks_completed / tasks_attempted. Range 0.0-1.0. Partial completions count as failed. Minimum for standard insurance underwriting: 0.70.
AS	Autonomy Score	1 - (H_assist / H_op). Range 0.0-1.0. Derived field — must equal 1 minus intervention rate. Target for standard insurance rates: >= 0.85.
IR	Intervention Rate	H_assist / H_op. Complement of AS. The raw operational measurement from which AS is derived.
MTBI	Mean Time Between Incidents	H_op / incident_count. Incidents include unplanned stoppages, safety triggers, task failures requiring restart, and human interventions. Analogous to MTBF in classical reliability engineering.
HLE	Human Labor Equivalent	Fully-loaded annual cost of the human worker(s) the robot replaces or augments. Formula: (hourly_wage x 1.35) x H_op. The 1.35 multiplier is the standard employer burden factor covering payroll taxes, benefits, workers compensation, and overhead.
DF	Displacement Factor	Degree of human labor displacement. 1.0 = full replacement of one worker. 0.5 = augmentation — robot and human each doing half the work. 0.0 = entirely new capability, no displacement. Operator-provided.
R	Revenue Direct	Direct revenue attributable to robot in period. For RaaS: contracted value delivered. For internal deployments: typically 0, with value captured via HLE displacement.
TCO	Total Cost of Ownership	Annual: (P/L) + C_energy + C_maint + C_operator + C_insurance + C_integration. Each component defined in Section 4.
MRS	MARS Risk Score	Composite insurance underwriting input. Formula defined in Section 6. Range 0.0-1.0. Higher = lower risk.

3. The AM-ROI Formula

3.1 Master Formula

The Autonomous Machine ROI (AM-ROI) is the canonical MARS performance metric. It expresses the economic return generated by a robot relative to its full cost of ownership, normalised against human labour as the baseline comparison.

```
AM-ROI  =  [ (R + HLE x DF) x TSR x AS ]  /  TCO_annual
```

Where TCO_annual = (P/L) + C_energy + C_maint + C_operator + C_insurance + C_integration

3.2 Interpreting the Result

AM-ROI Value	Economic Meaning	Operational Implication
< 1.0	Below break-even	Robot costs more per year than the human labour it displaces. Normal in Year 1 for current-generation humanoids as TSR and AS are still maturing. Not necessarily a failure — trajectory matters.
1.0	Parity	Robot costs exactly as much as equivalent human labour. No economic gain yet, but automation benefits (consistency, safety, scalability) may still justify deployment.
1.0 - 1.4	Marginal positive	Robot generating modest return above labour baseline. Acceptable for Year 2 deployments with improving trajectory.
>= 1.4	Production threshold	MARS minimum target for scaled deployment decisions. Robot delivering meaningful economic return. Standard insurance rates apply.
>= 2.0	Strong ROI	Robot significantly outperforming human labour baseline. Typical for well-deployed cobots and AMRs in Year 3+.

3.3 TCO Component Breakdown

Every cost component must be included for the AM-ROI to be meaningful. Omitting components inflates ROI and produces misleading comparisons.

Component	Formula	Typical Range (2026 Market)
Capital (P/L)	`Purchase price / lifespan years`	$15,000-$35,000/yr for $150K-$200K humanoid over 5-7 yr lifespan
Energy (C_energy)	`kWh_annual x local_rate`	$800-$2,500/yr for humanoid on 10hr daily shift
Maintenance (C_maint)	`P x maintenance_rate`	$8,000-$22,500/yr (8-15% of P annually for humanoids)
Operator (C_operator)	`Fleet manager cost / robots managed`	$3,000-$12,000/yr (ratio 1:10 to 1:250 robots per manager)

Insurance (C_insurance)	`P x insurance_rate`	$1,500-$5,250/yr (1-3.5% of P — emerging market, wide variance)
Integration (C_integration)	`Platform + SDK annual cost / robots`	$3,000-$15,000/yr (RobOps platform + MARS integration)

NOTE: *All cost ranges above are informed estimates based on publicly available market data as of early 2026. No large-scale published cost studies for humanoid robot deployments exist yet. These figures should be replaced with operator actuals wherever available.*

4. Worked Examples

4.1 Example A — Agility Digit at Toyota TMMC (Canada)

Background

On February 19, 2026, Agility Robotics and Toyota Motor Manufacturing Canada (TMMC) announced a commercial Robot-as-a-Service agreement, the first confirmed large-scale humanoid RaaS deployment in North America. Digit handles parts bin movement and conveyor loading on a 10-hour daily shift, 250 operating days per year.

This example models Year 1 performance at conservative but realistic parameters, then projects the trajectory to break-even.

Inputs

Input	Value	Source / Basis
Purchase price (P)	C$150,000	Forbes estimate based on CEO payback comments
Expected lifespan (L)	5 years	Conservative for current-generation humanoid
Operating hours (H_op)	2,500 hrs/yr	10hr shift x 250 days
Autonomy Score (AS) — Yr 1	0.90	MARS automotive logistics benchmark, Year 1
Task Success Ratio (TSR) — Yr 1	0.88	MARS automotive logistics benchmark, Year 1
Displacement Factor (DF)	0.85	Partial — human handles complex exceptions
Human Labour Equivalent (HLE)	C$72,000/yr	Canadian manufacturing worker, fully loaded (base C$26/hr x 1.35 x 2,050 hrs)
Energy cost (C_energy)	C$1,200/yr	2 kWh/shift x 250 days x C$0.12/kWh x 2 charges
Maintenance (C_maint)	C$15,000/yr	10% of P annually — Year 1 estimate
Operator allocation (C_operator)	C$6,000/yr	Fleet manager at 1:25 ratio
Insurance (C_insurance)	C$3,000/yr	2% of P — emerging market rate
Integration (C_integration)	C$5,000/yr	Arc platform + MARS integration

Calculation — Year 1

```
TCO_annual = (C$150,000 / 5) + C$1,200 + C$15,000 + C$6,000 + C$3,000 +
C$5,000
           = C$30,000 + C$30,200
           = C$60,200 / year

Net Value  = (R + HLE x DF) x TSR x AS
```

```
            = (0 + C$72,000 x 0.85) x 0.88 x 0.90
            = C$61,200 x 0.88 x 0.90
            = C$48,470 / year

AM-ROI Year 1  = C$48,470 / C$60,200  =  0.80
```

Trajectory Projection

Year	TSR	AS	AM-ROI	Notes
1	0.88	0.90	0.80	Below break-even — normal for Year 1 humanoid. Software still maturing.
2	0.91	0.93	1.08	Crosses parity. Software updates and operator learning improving performance.
3	0.93	0.95	1.35	Approaching production threshold. Fleet manager efficiency gains kicking in.
5	0.95	0.96	1.58	Strong ROI. Maintenance rate may increase — monitor C_maint.

Interpretation: AM-ROI of 0.80 in Year 1 is below economic break-even but consistent with Agility Robotics CEO Peggy Johnson's public statement that Digit pays for itself in approximately 2 years — which requires TSR improving to ~0.92 and AS to ~0.95, achievable with software updates. MARS makes this improvement trajectory **trackable and auditable** — exactly what insurers need to structure performance-improving coverage terms.

4.2 Example B — Figure 02 vs Boston Dynamics Atlas (BMW Spartanburg)

Background

BMW Plant Spartanburg in South Carolina simultaneously deployed both Figure 02 and Boston Dynamics Atlas units on the same precision assembly production line in early 2026. This created the first confirmed head-to-head humanoid comparison in a live production environment.

Without MARS: a CFO asking 'which robot should we scale?' has no neutral answer. With MARS: the answer is a 30-second dashboard query.

Comparison

Metric	Figure 02	Atlas Commercial
Purchase price (P)	US$175,000	US$200,000
Expected lifespan (L)	7 years	7 years
Annual capital cost (P/L)	US$25,000	US$28,571
Task Success Ratio (TSR)	0.91	0.89
Autonomy Score (AS)	0.92	0.94
Maintenance rate	10% of P = US$17,500/yr	12% of P = US$24,000/yr
Energy cost	US$1,400/yr	US$1,600/yr
Operator + Insurance + Integration	US$12,000/yr	US$12,000/yr
TCO Annual	US$55,900/yr	US$66,171/yr
HLE displaced (German automotive)	EUR 120,000/yr (~US$130,000)	EUR 120,000/yr (~US$130,000)
Net Value	US$100,490/yr	US$100,565/yr
AM-ROI	1.80	1.52
Estimated payback period	2.9 years	3.8 years

MARS Insight

Figure 02 delivers **18% better AM-ROI** than Atlas in this scenario. Performance output is nearly identical — TSR and AS are comparable — but Figure 02 wins decisively on economics because its lower purchase price and lower maintenance rate produce a significantly better TCO.

The BMW procurement team cannot currently make this calculation because no neutral cross-manufacturer standard exists. MARS makes it a 30-second dashboard query. This intelligence justifies scaled deployment decisions worth hundreds of millions of dollars.

NOTE: *Both purchase prices and performance figures above are estimates based on publicly available information as of early 2026. BMW has not published official performance data for either deployment. These figures are illustrative of MARS methodology, not certified benchmarks.*

5. Performance Benchmarks by Deployment Type

These benchmarks define what 'good' looks like for each deployment environment. They are used in: (1) MTBI normalisation for the MRS formula; (2) deployment-type context for AM-ROI interpretation; (3) RaaS SLA threshold guidance.

NOTE: *All benchmarks below are v0.9 estimates. They are not derived from a published empirical dataset — no such dataset exists for humanoid robots at production scale as of March 2026. They are based on available performance data from early deployments, manufacturer specifications, and analogies to cobot and AMR performance literature. MARS explicitly invites challenge and refinement from the robotics community.*

Deployment Type	TSR Range	AS Range	MTBI Benchmark	Min AM-ROI	Insurance Tier
Automotive manufacturing	0.88-0.95	0.90-0.97	200+ hrs	>= 1.8	Standard
Logistics / warehouse	0.85-0.93	0.88-0.95	150+ hrs	>= 1.6	Standard
Healthcare	0.92-0.98	0.80-0.90	500+ hrs	>= 1.4	Elevated
Retail	0.80-0.90	0.85-0.92	100+ hrs	>= 1.3	Standard
Agricultural	0.75-0.88	0.80-0.90	80+ hrs	>= 1.2	Standard
Last-mile delivery	0.82-0.91	0.88-0.95	120+ hrs	>= 1.5	Elevated
Construction	0.70-0.85	0.75-0.88	60+ hrs	>= 1.1	Elevated

6. Insurance Framework

6.1 MARS Risk Score (MRS)

The MARS Risk Score is the canonical insurance underwriting input produced by MARS. It is a single normalised value computed from four weighted performance dimensions, designed as a direct actuarial input for premium calculation.

```
MRS = (TSR x 0.30) + (AS x 0.25) + (MTBI_norm x 0.20) +
(Safety_norm x 0.25)
```

Where:

- MTBI_norm = min(MTBI_actual / MTBI_benchmark_for_deployment_type, 1.0) — capped at 1.0 so exceeding benchmark does not inflate score
- Safety_norm = safety_score / 100 — normalises the 0-100 safety_score to the 0.0-1.0 scale required by the formula
- Weights sum to 1.00. v0.9 defaults. Insurers may apply contract-specific weights.

NOTE: *The MRS formula weights (0.30, 0.25, 0.20, 0.25) were constructed based on logical priority — task performance, autonomy quality, incident frequency, and physical safety. They have NOT been validated against actuarial loss data. A qualified actuary should review and adjust weights before use in binding insurance contracts.*

6.2 MRS Premium Tiers

MRS Range	Risk Class	Annual Premium Rate	Coverage Terms	Data Requirement
0.90-1.00	Excellent	0.8-1.2% of P	Full coverage, standard deductible	90 days live data
0.80-0.89	Good	1.2-2.0% of P	Full coverage, moderate deductible	60 days live data
0.70-0.79	Standard	2.0-3.5% of P	Limited coverage, high deductible	30 days live data
0.60-0.69	Elevated	3.5-6.0% of P	Conditional / pilot terms only	Simulation data accepted
< 0.60	Uninsurable	Not eligible	Remediation plan required first	N/A

Every 0.10 improvement in MRS reduces annual premium by approximately 0.8-1.2% of purchase price. For a US$175,000 Figure 02, improving MRS from 0.75 to 0.85 saves approximately US$1,400-$2,100 per year in insurance premium. This creates a direct financial incentive for operators to improve robot performance — aligning operator, insurer, and manufacturer interests around a neutral data standard.

6.3 RaaS Parametric SLA Triggers

For Robot-as-a-Service deployments, MARS data enables parametric insurance products — coverage that triggers automatically when performance falls below contracted SLA thresholds, without requiring human claims investigation.

Metric	Standard Threshold	Breach Duration	Automatic Trigger
Task Success Ratio (TSR)	< 0.85	7 consecutive days	Daily rate credit to operator
Uptime	< 0.90 (90%) in rolling 30 days	Any 30-day window	Pro-rata RaaS fee refund
Autonomy Score (AS)	< 0.80 on 14-day rolling avg	14 consecutive days	Labour cost differential coverage
MTBI	< 100 hrs on 30-day rolling avg	Any 30-day window	Enhanced maintenance coverage activates
Vision accuracy	< 0.92 on 30-day rolling avg	Any 30-day window	AI performance review clause triggers
Collision index	> 2.0 per 100 hrs in 30-day window	Any 30-day window	Liability coverage enhancement

NOTE: *Vision accuracy threshold of 0.92 reflects actual achievable performance for current-generation humanoids in structured production environments (controlled lighting, known object sets). In unstructured environments, achievable accuracy drops to 0.75-0.86. SLA contracts must specify the operational conditions under which measurement applies.*

7. Depreciation and Lifecycle Valuation

7.1 Why Standard Straight-Line is Wrong for Robots

Standard straight-line depreciation (equal loss of value each year) is inaccurate for autonomous machines. Humanoid robots face high early obsolescence risk as the technology moves rapidly, then stabilise into a predictable wear cycle, before declining slowly as hardware outlives software support. MARS defines a three-phase model that reflects this reality.

7.2 MARS-DEP Three-Phase Depreciation Model

Phase	Period	Annual Rate	Primary Driver
Phase 1: Rapid Initial	Year 1-2 (days 0-730)	20-25% of P/yr	Technology obsolescence risk — next generation robot may arrive. Software immaturity means performance is still improving, reducing current unit value.
Phase 2: Stable Operational	Year 3-6 (days 731-2190)	10-15% of P/yr	Normal hardware wear. Predictable component replacement cycle. Performance stable.
Phase 3: Terminal	Year 7+ (days 2191+)	5-8% of P/yr	Residual hardware value. Parts availability declining. Manufacturer software support may end. Resale market thin.

7.3 Resale Value Estimation

MARS creates the first auditable performance provenance record for autonomous machines, enabling a substantiated resale value calculation analogous to a vehicle service history.

```
RSV = P x (1 - cumulative_depreciation_rate)
        x condition_factor
        x model_factor

Where:
  condition_factor = (TSR_lifetime_avg x 0.6) + (AS_lifetime_avg x 0.4)
  model_factor     = 0.5 to 1.0 based on manufacturer support status
```

Why weighted average, not product: Using TSR x AS as a product over-penalises. If TSR = 0.88 and AS = 0.90, the product gives 0.79 (21% discount) but a weighted average gives 0.888 (11% discount). A weighted average better reflects how the used robot market actually prices performance — both dimensions matter but neither alone kills the value.

NOTE: *The resale market for humanoid robots is nascent as of 2026. No reliable market comparables exist. RSV estimates should be treated as indicative only until a functioning secondary market develops.*

8. Implementation Guide

8.1 Minimum Data Requirements

Tier	Fields	Enables
Core (Required)	`machine_id, vendor, model, timestamp, uptime_hours, downtime_hours, incident_count, tasks_completed, tasks_attempted, H_auto, H_op`	MTBI, TSR, AS, intervention_rate calculation. Basic operational record.
Insurance (Required)	`falls_detected, collision_events, near_miss_events, ISO compliance flags, safety_score`	Full MARS Risk Score (MRS). Insurance underwriting eligibility.
Financial (Operator-provided)	`purchase_price, expected_lifespan_years, hle_annual, displacement_factor, deployment_date`	Complete AM-ROI calculation. TCO breakdown. Payback period.
AI Performance (Recommended)	`vision_accuracy, drift_index, hallucination_rate, model_version`	AI-specific insurance (aiSure-type). Drift monitoring. Model version tracking.
Energy / Maintenance (Recommended)	`energy_kwh_total, maintenance_cost, components_replaced, battery_health_pct`	Full TCO accuracy. Predictive maintenance. Asset lifecycle tracking.

8.2 Integration Pathways

Pathway	Complexity	Time	Best For
ROS2 Topic Bridge	Low-Medium	1-2 weeks	Any ROS2-compatible autonomous machine. Subscribe to standard topics and map to MARS fields.
Vendor Cloud API	Low	1 week	Robots with existing cloud telemetry (Figure, Boston Dynamics, Agility). Map vendor fields to MARS schema.
MARS SDK (Direct)	Medium	2-4 weeks	Greenfield deployments. Full control over data collection. Most complete MARS record.
InOrbit / Viam Integration	Low	Days	Fleets already on RobOps platforms. MARS as an analytics layer above existing telemetry.

Isaac Sim / Gazebo	Low	Days	Pilot validation and demo environments. Generate synthetic MARS-compliant records.
Manual Entry Baseline	Minimal	Hours	Proof of concept. Historical data import. Operator-provided financial inputs only.

8.3 4-Week Pilot Programme

Week	Phase	Activities and Deliverables
Week 1	Integration and Baseline	Connect robot via chosen integration pathway. Activate Core tier logging. Establish TSR, AS, and MTBI baselines. Confirm SHA-256 audit hash chain integrity.
Week 2	Safety and Stability	Activate Insurance tier. Capture falls, collisions, near-misses. Log ISO compliance indicators. Establish fall severity index baseline.
Week 3	AI Performance and Economics	Activate vision accuracy and drift monitoring. Inject operator Financial tier inputs (purchase price, HLE, DF). Generate first AM-ROI calculation.
Week 4	Reporting and Validation	Generate full AM-ROI report. Compute MRS score. Produce insurer-ready data package. Validate RaaS SLA compliance status if applicable.

9. Confidence Levels — What to Trust

This section exists because MARS is an honest standard. Every element below is classified by the strength of the evidence behind it. Readers should apply appropriate scrutiny to estimated values.

Element	Confidence	Basis and Caveats
AM-ROI formula structure	High	Standard capital investment ROI methodology adapted from industrial asset economics. Formula logic is sound.
TCO component categories	High	These cost categories are standard in industrial asset management literature and enterprise robotics procurement.
ISO standard references (10218, TS 15066, 13482)	High	Verified against ISO published standards catalogue. iso_25785 was removed in v0.9 — it does not exist.
MTBI methodology	High	Directly analogous to MTBF in reliability engineering (IEC 60050-192). Adapted terminology for software and intervention events.
TSR and AS definitions	High	Mathematically clean, internally consistent. AS = 1 - IR is an enforced constraint, not an estimate.
Depreciation phase model	Medium	Analogised from technology equipment and early cobot market data. No humanoid-specific published depreciation studies exist. Rates are informed estimates.
Benchmark TSR/AS ranges by deployment type	Medium-Low	Based on available early deployment data and manufacturer specifications. No large-scale published empirical dataset exists. Explicitly invite challenge.
MRS weighting factors	Medium-Low	Constructed on logical priority. Not derived from actuarial loss data. Must be reviewed by a qualified actuary before use in binding contracts.
Insurance premium rate ranges (% of P)	Low-Medium	Based on emerging market observations and analogies to other equipment insurance. Robot insurance market is immature and pricing is highly variable.
Vision accuracy threshold (0.92)	Medium	Consistent with Boston Dynamics published benchmark ranges (0.88-0.94) in controlled environments. Drops significantly in unstructured settings.
Cost benchmarks (maintenance %, energy, operator ratios)	Medium	Based on publicly available market data and IFR World Robotics Report analogies. Should be replaced with operator actuals wherever possible.

10. Standard Roadmap

Version	Target	Milestone
v0.9 (current)	March 2026	Core AM-ROI formula. 19-object schema. MARS Risk Score. RaaS SLA triggers. 3-phase depreciation. Public draft open for community review.
v1.0	Q2 2026	Community-validated formula. ROS2 SDK published. First live pilot data. Insurer design partner endorsement. GitHub repository active.
v1.5	Q3 2026	Multi-robot fleet ROI aggregation. Cross-manufacturer benchmarking dashboard. RaaS SLA automation via parametric triggers.
v2.0	Q1 2027	Agricultural module. Healthcare module. Parametric insurance product templates. AI drift insurance product definition.
v3.0	2028	Standards body submission (ISO, IFR). Regulatory framework alignment (EU AI Act, Singapore MGF). Industry certification programme.

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